Design System Drift Audit: vercel/ai-chatbot (DRAFT)

DRAFT for review. Not for the client until it has been reviewed and edited. Every class below is proposed by rule and confirmed by hand before the report leaves.

Run: vercel/ai-chatbot at c2f8235, against @/components/ui@0.0.0. 124 files checked in 0.5 seconds by @zenginui/cli@0.9.0, 2026-10-15.

Summary

- This run reported 109 findings in 82 groups. A group is one rule and one found value with one proposed class: one decision.
- Hand review covers 109 findings in 82 groups, up to the cap of 150, taken in rank order (severity times the number of files). 0 findings are summarised, not classified by hand. 109 + 0 = 109.
- False positives proposed in the hand-review set: 0 of 109. The rate is stated once each one has been confirmed by hand.
- Every number here comes from this run and is reconciled above. No figure from another run is quoted.

```
Proposed class | Findings | Groups
True drift | 75 | 56
Needs a decision | 34 | 26
Intentional exception | 0 | 0
False positive | 0 | 0
Hand-review set | 109 | 82
```

Top issues

G01: classname-policy rounded-full (Needs a decision)

- What: className on Button may not set border-radius. border-radius is owned by the component; use a variant or a token.
- Spread: 4 findings in 3 files, at warn. Involves Button.
- Why this class: True by the component's contract, reported at warn: whether className.allow should widen for this component is a policy call.
- Fix: No fix the engine can write; decided by hand.
- Examples: F001 components/ai-elements/conversation.tsx:88, F002 components/ai-elements/conversation.tsx:155, F003 components/ai-elements/suggestion.tsx:48.
- Findings: F001, F002, F003, F004.

G02: color-literal text-red-500 (True drift)

- What: text-red-500 references the Tailwind default palette (--color-red-500), not a system token.
- Spread: 4 findings in 2 files, at error. Involves color.destructive.
- Why this class: color-literal at error, with a nearest fix.
- Fix: text-destructive (nearest). destructive is the system's token for this role. Chosen by role, not by value; verify it reads correctly here.
- Examples: F005 components/chat/message.tsx:264, F006 components/chat/message.tsx:287, F007 components/chat/message.tsx:323.
- Findings: F005, F006, F007, F008.

G03: color-literal text-white (True drift)

- What: text-white references the Tailwind default palette (--color-white), not a system token.
- Spread: 3 findings in 2 files, at error. Involves color.card.
- Why this class: color-literal at error, with a nearest fix.
- Fix: text-card (nearest). Several tokens share this value. Pick by role: color.card, color.popover.
- Examples: F009 components/chat/preview-attachment.tsx:47, F010 components/chat/weather.tsx:363, F011 components/chat/weather.tsx:412.
- Findings: F009, F010, F011.

G04: color-literal bg-black/80 (True drift)

- What: bg-black/80 references the Tailwind default palette (--color-black), not a system token.
- Spread: 2 findings in 2 files, at error. Involves color.foreground.
- Why this class: color-literal at error, with a nearest fix.
- Fix: bg-foreground (nearest). Nearest by value: color.foreground = #060606. Verify the semantic role before applying.
- Examples: F012 components/ui/alert-dialog.tsx:39, F013 components/ui/dialog.tsx:42.
- Findings: F012, F013.

G05: classname-policy gap-1.5 (Needs a decision)

- What: className on Badge may not set gap. className on Badge is limited to: margin, flex-item, grid-item, position, display.
- Spread: 2 findings in 2 files, at warn. Involves Badge.
- Why this class: True by the component's contract, reported at warn: whether className.allow should widen for this component is a policy call.
- Fix: No fix the engine can write; decided by hand.
- Examples: F014 components/ai-elements/tool.tsx:69, F015 components/chat/visibility-selector.tsx:117.
- Findings: F014, F015.

G06: classname-policy px-4 (Needs a decision)

- What: className on Button may not set padding-inline. padding is owned by the size prop.
- Spread: 2 findings in 2 files, at warn. Involves Button.
- Why this class: True by the component's contract, reported at warn: whether className.allow should widen for this component is a policy call.
- Fix: No fix the engine can write; decided by hand.
- Examples: F016 components/ai-elements/suggestion.tsx:48, F017 components/chat/chat-header.tsx:55.
- Findings: F016, F017.

G07: classname-policy rounded-lg (Needs a decision)

- What: className on Button may not set border-radius. border-radius is owned by the component; use a variant or a token.
- Spread: 2 findings in 2 files, at warn. Involves Button.

- Why this class: True by the component's contract, reported at warn: whether className.allow should widen for this component is a policy call.
- Fix: No fix the engine can write; decided by hand.
- Examples: F018 components/chat/chat-header.tsx:55, F019 components/chat/visibility-selector.tsx:117.
- Findings: F018, F019.

G08: color-literal text-white/60 (True drift)

- What: text-white/60 references the Tailwind default palette (--color-white), not a system token.
- Spread: 3 findings in 1 file, at error. Involves color.card.
- Why this class: color-literal at error, with a nearest fix.
- Fix: text-card (nearest). Several tokens share this value. Pick by role: color.card, color.popover.
- Examples: F020 components/chat/weather.tsx:348, F021 components/chat/weather.tsx:404, F022 components/chat/weather.tsx:421.
- Findings: F020, F021, F022.

G09: color-literal text-white/80 (True drift)

- What: text-white/80 references the Tailwind default palette (--color-white), not a system token.
- Spread: 3 findings in 1 file, at error. Involves color.card.
- Why this class: color-literal at error, with a nearest fix.
- Fix: text-card (nearest). Several tokens share this value. Pick by role: color.card, color.popover.
- Examples: F023 components/chat/weather.tsx:347, F024 components/chat/weather.tsx:365, F025 components/chat/weather.tsx:380.
- Findings: F023, F024, F025.

G10: color-literal bg-red-50 (True drift)

- What: bg-red-50 references the Tailwind default palette (--color-red-50), not a system token.
- Spread: 2 findings in 1 file, at error. Involves color.destructive.
- Why this class: color-literal at error, with a nearest fix.
- Fix: bg-destructive (nearest). destructive is the system's token for this role. Chosen by role, not by value; verify it reads correctly here.
- Examples: F026 components/chat/message.tsx:264, F027 components/chat/message.tsx:287.
- Findings: F026, F027.

Fix path by class

True drift

Fix in code, in group order. Apply exact fixes first; check each nearest fix by eye, since a neighbouring step can be a different design.

```
Group | Rule | Found | Findings | Files
G02 | color-literal | text-red-500 | F005, F006, F007, F008 | 2
G03 | color-literal | text-white | F009, F010, F011 | 2
G04 | color-literal | bg-black/80 | F012, F013 | 2
```

G08 | color-literal | text-white/60 | F020, F021, F022 | 1
G09 | color-literal | text-white/80 | F023, F024, F025 | 1
G10 | color-literal | bg-red-50 | F026, F027 | 1
G11 | color-literal | bg-white/10 | F028, F029 | 1
G12 | color-literal | border-red-200 | F030, F031 | 1
G13 | color-literal | dark:bg-red-950/50 | F032, F033 | 1
G14 | color-literal | dark:text-neutral-50 | F034, F035 | 1
G15 | color-literal | text-blue-200 | F036, F037 | 1
G16 | color-literal | text-white/70 | F038, F039 | 1
G17 | color-literal | text-white/90 | F040, F041 | 1
G18 | color-literal | text-yellow-200 | F042, F043 | 1
G19 | color-literal | bg-amber-500 | F044 | 1
G20 | color-literal | bg-black/20 | F045 | 1
G21 | color-literal | bg-black/40 | F046 | 1
G22 | color-literal | bg-black/50 | F047 | 1
G23 | color-literal | bg-black/60 | F048 | 1
G24 | color-literal | bg-white/20 | F049 | 1
G25 | color-literal | dark:bg-neutral-900 | F050 | 1
G26 | color-literal | dark:bg-neutral-950 | F051 | 1
G27 | color-literal | data-[type=error]:text-red-600 | F052 | 1
G28 | color-literal | data-[type=success]:text-green-600 | F053 | 1
G29 | color-literal | hover:bg-black/80 | F054 | 1
G30 | color-literal | text-blue-600 | F055 | 1
G31 | color-literal | text-emerald-500 | F056 | 1
G32 | color-literal | text-green-600 | F057 | 1
G33 | color-literal | text-neutral-500 | F058 | 1
G34 | color-literal | text-orange-600 | F059 | 1
G35 | color-literal | text-red-400 | F060 | 1
G36 | color-literal | text-red-600 | F061 | 1
G37 | color-literal | text-yellow-600 | F062 | 1
G38 | component-substitution | <button className="flex size-6 cursor-pointer items-center justify-center rounde | F063 | 1
G39 | component-substitution | <button className="flex size-7 items-center justify-center rounded-md text-muted | F064 | 1
G40 | component-substitution | <button className="flex size-7 items-center justify-center rounded-md text-muted | F065 | 1
G41 | component-substitution | <button className="rounded px-1.5 py-0.5 text-muted-foreground/50 transition-col | F066 | 1
G42 | component-substitution | <button className="absolute top-1.5 right-1.5 flex size-5 items-center justify-c | F067 | 1
G43 | component-substitution | <button className="flex w-full items-center rounded-2xl border border-border/30 | F068 | 1
G44 | component-substitution | <button className="inline-flex items-center justify-center gap-2 rounded-lg bg-f | F069 | 1
G45 | component-substitution | <button className="inline-flex items-center justify-center rounded-lg border bor | F070 | 1
G46 | component-substitution | <button className={cn("flex items-center justify-center rounded-full p-3 text-m | F071 | 1
G47 | component-substitution | <button className={cn("flex size-7 items-center justify-center rounded-md text- | F072 | 1
G48 | component-substitution | <button aria-label="Scroll to bottom" className={'absolute bottom-4 left-1/2 z-1 | F073 | 1
G49 | component-substitution | <button className="rounded-md bg-primary px-3 py-1.5 text-primary-foreground tex | F074 | 1
G50 | component-substitution | <button className="rounded-md px-3 py-1.5 text-muted-foreground text-sm

transiti | F075 | 1
G51 | component-substitution | <button className="cursor pointer flex w-fit flex-row items-start justify-betwee | F076 | 1
G52 | component-substitution | <button className="flex w-fit cursor-pointer flex-row items-center gap-2 rounded | F077 | 1
G53 | component-substitution | <button className="group flex size-8 items-center justify-center rounded-lg bord | F078 | 1
G54 | component-substitution | <button className="rounded-xl border border-border/30 bg-card/20 px-3 py-2.5 tex | F079 | 1
G55 | component-substitution | <button className={cn("flex w-full items-center gap-3 px-4 py-2.5 text-left tra | F080 | 1
G56 | component-substitution | <button className="w-full px-1 py-0.5 text-left text-red-500" type="submit"> | F081 | 1
G57 | spacing-literal | has-[>kbd]:ml-[-0.15rem] | F082 | 1
G58 | spacing-literal | has-[>kbd]:mr-[-0.15rem] | F083 | 1
G59 | token-reference | toast-mobile:w-[356px] | F084 | 1
G60 | token-reference | toast-mobile:w-fit | F085 | 1

Needs a decision

A decision for the team before any code changes: add the missing token, name the file as a foundation, or widen the component's className.allow. Each decision closes a whole group.

Group | Rule | Found | Findings | Files
G01 | classname-policy | rounded-full | F001, F002, F003, F004 | 3
G05 | classname-policy | gap-1.5 | F014, F015 | 2
G06 | classname-policy | px-4 | F016, F017 | 2
G07 | classname-policy | rounded-lg | F018, F019 | 2
G61 | classname-policy | dark:bg-background | F086, F087 | 1
G62 | classname-policy | dark:hover:bg-muted | F088, F089 | 1
G63 | classname-policy | [&>*:not(:first-child)]:rounded-l-md | F090 | 1
G64 | classname-policy | [&>*:not(:last-child)]:rounded-r-md | F091 | 1
G65 | classname-policy | **:data-[slot=command-input-wrapper]:h-auto | F092 | 1
G66 | classname-policy | active:translate-y-0 | F093 | 1
G67 | classname-policy | bg-foreground | F094 | 1
G68 | classname-policy | border | F095 | 1
G69 | classname-policy | border-border/50 | F096 | 1
G70 | classname-policy | cursor-pointer | F097 | 1
G71 | classname-policy | focus-visible:border-border/50 | F098 | 1
G72 | classname-policy | focus-visible:ring-0 | F099 | 1
G73 | classname-policy | hover:bg-foreground/90 | F100 | 1
G74 | classname-policy | hover:text-foreground | F101 | 1
G75 | classname-policy | rounded-md | F102 | 1
G76 | classname-policy | shadow-none | F103 | 1
G77 | classname-policy | text-background | F104 | 1
G78 | classname-policy | text-muted-foreground | F105 | 1
G79 | classname-policy | text-xs | F106 | 1
G80 | classname-policy | transition-colors | F107 | 1
G81 | classname-policy | translate-x-[-50%] | F108 | 1
G82 | classname-policy | whitespace-nowrap | F109 | 1

Intentional exception

Keep it and write the reason after the zengin-allow comment's colon, so the engine honours it and the next reader knows why; or remove the exception.

No group in the hand-review set.

False positive

Not a code change. Tune the definitions (a passthrough type, a foundation file) so the engine stops reporting it, and count it in the report's false-positive rate.

No group in the hand-review set.

Beyond the hand-review cap

Every finding is in the hand-review set; the cap of 150 was not reached.

What the scan cannot see

- Drift outside the engine's seven rule kinds (color-literal, spacing-literal, token-reference, unknown-prop, unknown-prop-value, classname-policy, component-substitution): behavioural, architectural and most accessibility drift.
- What renders at runtime: class names built from runtime values, styles set from JavaScript, and the visual result of an override.
- Files outside the configured scope, and any system the definitions do not describe.
- Whether a finding matters to a user. That judgement is the hand review.

Evidence

Every finding id above resolves in findings.json, with its file:line, rule, found value, the token or component involved, its proposed class and the reason for it. run.json records the run.